

2016.0 RANGE ROVER (LG), 501-20

SUPPLEMENTAL RESTRAINT SYSTEM

DIAGNOSIS AND TESTING

PRINCIPLE OF OPERATION

For a detailed description of the supplemental restraints system and operation, refer to the relevant Description and Operation section in the workshop manual. REFER to: (501-20B Supplemental Restraint System)

[Airbag and Safety Belt Pretensioner Supplemental Restraint System](#)

(Description and Operation),

[Airbag Supplemental Restraint System](#) (Diagnosis and Testing).

SAFETY INFORMATION

WARNINGS:

- TO AVOID ACCIDENTAL DEPLOYMENT AND POSSIBLE PERSONAL INJURY, THE BACKUP POWER SUPPLY MUST BE DEPLETED BEFORE REPAIRING OR REPLACING ANY AIRBAG SUPPLEMENTAL RESTRAINT SYSTEM (SRS) COMPONENTS. TO DEplete THE BACKUP POWER SUPPLY ENERGY, DISCONNECT THE BATTERY GROUND CABLE AND WAIT ONE MINUTE. FAILURE TO FOLLOW THIS INSTRUCTION MAY RESULT IN PERSONAL INJURY
- Do not use a multimeter to probe the restraints control module. It is possible for the power from the meter battery to trigger the activation of the airbags. Failure to follow this instruction may result in personal injury

NOTES:

- It is advisable not to use a cellular phone or to have a cellular phone in close proximity when working on the restraints control module or associated systems
- Given the legal implications of a restraints system failure, harness repairs to airbag module circuits are not acceptable. Where the text refers to "REPAIR the circuit", this will normally mean the replacement of a harness

Power Supply Depletion

Before beginning any work on the restraints control module/supplemental restraints system or related components:

1. Remove the ignition key
2. Disconnect the battery leads, ground first

3. Wait 2 minutes for the power circuit to discharge

There are comprehensive instructions on the correct procedures for supplemental restraints system repairs in the manual. Refer to the relevant section of the workshop manual

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the warranty policy and procedures manual (section B1.2), or determine if any prior approval programme is in operation, prior to the installation of a new module/component
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests

1. Verify the customer concern

1. Confirm the function of the warning lamp (if the warning lamp is inoperative, system faults will be signalled by an audible chime)

2. Visually inspect for obvious signs of damage and system integrity

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Check for the installation of non-standard accessories which may affect or obstruct the function of the supplemental restraint system ▪ Check the condition of trim, etc at the supplemental restraint system components ▪ Sensor(s) ▪ Pretensioner(s) ▪ Airbag module(s) ▪ Occupant classification sensor control module ▪ Seat position sensor	▪ Fuse(s) ▪ Wiring harness ▪ Battery condition, state of charge ▪ Ensure all electrical connector(s) are engaged correctly on the airbag circuits ▪ Ensure the restraints control module is correctly installed ▪ Warning indicator ▪ Impact sensor(s) ▪ Buckle sensor(s) ▪ Pretensioner(s) ▪ Airbag module(s) ▪ Passenger airbag deactivation switch ▪ Passenger airbag deactivation warning indicator ▪ Occupant classification sensor control module ▪ Seat position sensor ▪ Clockspring

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

Faults in supplemental restraint system harnesses would normally mean replacement of the relevant harness. Supplemental restraint system harness repairs are not recommended

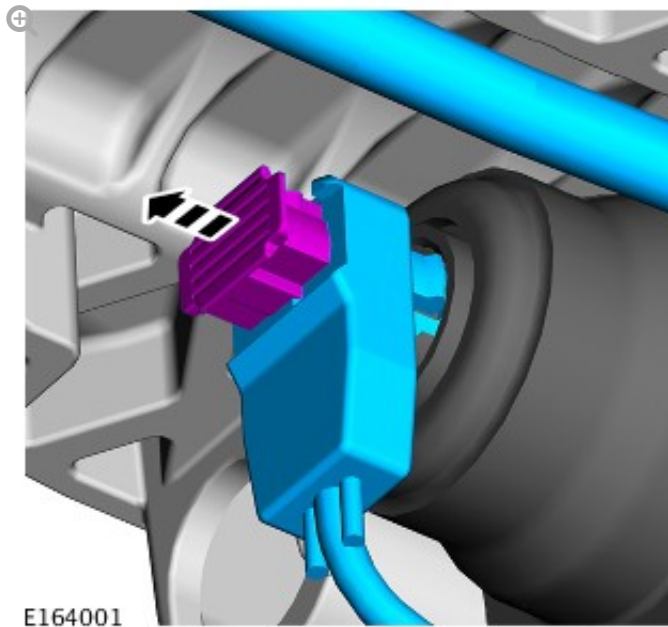
Symptom Chart

SYMPTOM	POSSIBLE CAUSES	ACTION
▪ Deployable device connector and/or wiring harness	▪ Wiring harness connector connected/disconnected without first disengaging the	

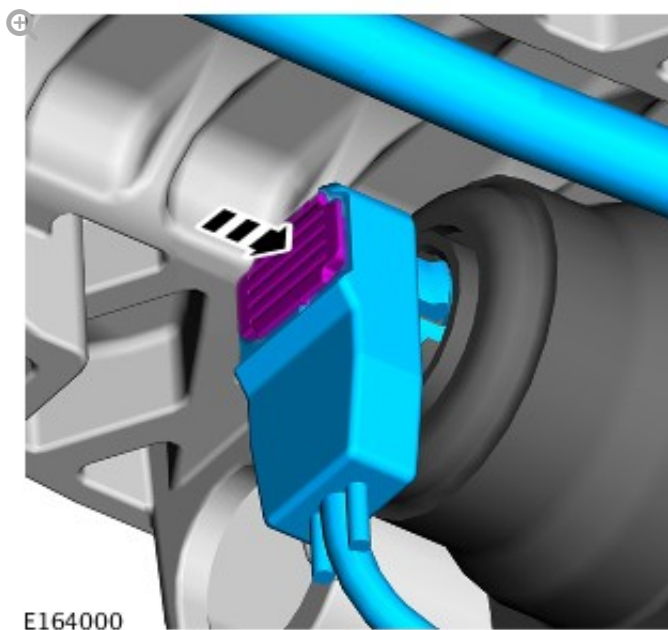
connector damaged	connector plunger	NOTE: Damage caused by an incorrect removal or installation procedure is not warrantable. Install a new wiring harness and/or deployable device as necessary
-------------------	-------------------	--

- If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required
- When installing or checking the supplemental Restraints System (SRS) with AK2 connectors the locking plunger must be engaged

SUPPLEMENTAL RESTRAINTS SYSTEM (SRS) AK2 TYPE CONNECTORS



- Incorrect Position (Disengaged)



- Correct Position (Engaged)

DTC INDEX

For a list of diagnostic trouble codes that could be logged on this vehicle, please refer to Section 100-00. REFER to: (100-00 General Information)

[Diagnostic Trouble Code Index - DTC: Occupant Classification Sensor](#)

[Control Module](#) (Description and Operation),

[Diagnostic Trouble Code Index - DTC: Restraints Control Module \(RCM\)](#)

(Description and Operation).